

Machine Translation Benchmarking

Evaluating NMT Systems for English → Hindi

Benchmark Evaluation Report

IITB Test Set · 4 Systems Compared

April 30, 2026

Outline

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- 2 Evaluation Metrics
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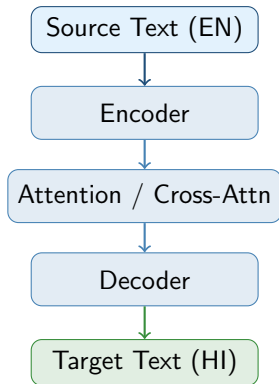
What is Machine Translation?

Definition

Machine Translation (MT) is the automated process of translating text from one natural language to another using computational models, without human intervention.

Key Paradigms:

- **Rule-Based MT (RBMT)** — hand-crafted linguistic rules
- **Statistical MT (SMT)** — phrase/word alignment probabilities
- **Neural MT (NMT)** — encoder-decoder deep learning



The English–Hindi Translation Challenge

Why English → Hindi?

- Hindi has **600M+ speakers** globally
- Significant demand in news, government, e-commerce
- Morphologically rich language with complex agreement
- **Low-resource** relative to European language pairs

Key Linguistic Challenges

- Script difference: Latin → Devanagari
- Free word order (SOV vs. SVO)
- Agglutinative morphology
- Named entity transliteration
- Code-mixing in source text
- Cultural / idiomatic expressions

Test Set: IIT Bombay English–Hindi Corpus

A widely used benchmark dataset containing aligned sentence pairs sourced from diverse domains including news, government documents, and general text.

Evaluation Metrics Overview

Lexical / N-gram Metrics

- **BLEU** (Bilingual Evaluation Understudy) — precision of n -grams vs. reference; ↑ higher is better
- **chrF++** — character n -gram F-score; more robust for morphology; ↑
- **TER** (Translation Edit Rate) — edit operations to reach reference; ↓ lower is better
- **METEOR** — alignment-based with synonymy; ↑

Neural / Semantic Metric

- **BERTScore-F1** — contextual embeddings from BERT; captures semantic similarity beyond surface form; ↑

Metric Limitations

No single metric perfectly captures translation quality. Human evaluation remains the gold standard, but automated metrics enable fast, reproducible comparison at scale.

BLEU Score in Depth

Formula:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right)$$

where p_n is modified n -gram precision and BP is the brevity penalty.

Score Interpretation Guide

BLEU Range	Quality Assessment
< 10	Poor, barely usable
10 – 19	Hard to understand
20 – 29	Gist conveyed
30 – 40	Understandable
40 – 50	Good

- Computed over a **corpus**, not per sentence
- Uses **clipped** counts to avoid over-generation
- Correlates moderately well with human judgement
- **Weaknesses:** ignores synonymy, word order, morphology
- Especially noisy for **morphologically rich** target languages like Hindi

Why Multiple Metrics?

chrF++ and BERTScore compensate

Systems Under Evaluation

[NMT] MarianMT (Helsinki-NLP)

- Encoder-decoder Transformer
- Trained on OPUS parallel corpora
- Lightweight, fast inference
- Widely used baseline in research

[NMT] mBART-50 (Meta AI)

- Multilingual BART pre-training
- Fine-tuned on 50 languages
- Denoising pre-training objective
- Strong cross-lingual transfer

[NMT] NLLB-600M (Meta AI)

- No Language Left Behind initiative
- 200-language multilingual model
- 600M parameter variant
- Dedicated low-resource language support

[API] Google Translate (API)

- Commercial neural MT system
- Massive proprietary training data
- Continuously updated model
- Industry gold-standard reference

Overall Metric Scores

Model	BLEU ↑	chrF++ ↑	TER ↓	METEOR ↑	BERTScore-F1 ↑
MarianMT (Helsinki)	35.50	30.72	83.15	28.56	78.92
mBART-50 (Meta)	48.12	41.53	77.91	40.60	83.86
NLLB-600M (Meta)	56.20	48.11	62.52	48.02	85.85
Google Translate	60.00	52.07	57.70	51.97	87.25

#1 Winner

●

Google Translate

Ranked 1st on all 5 metrics

#2 Runner-up

2nd

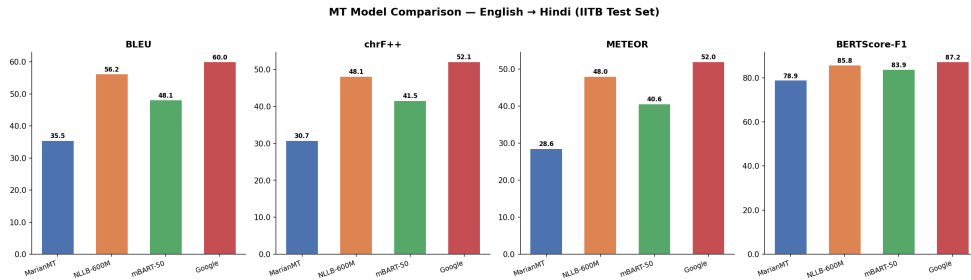
NLLB-600M

Strong open-source choice

Baseline Gap

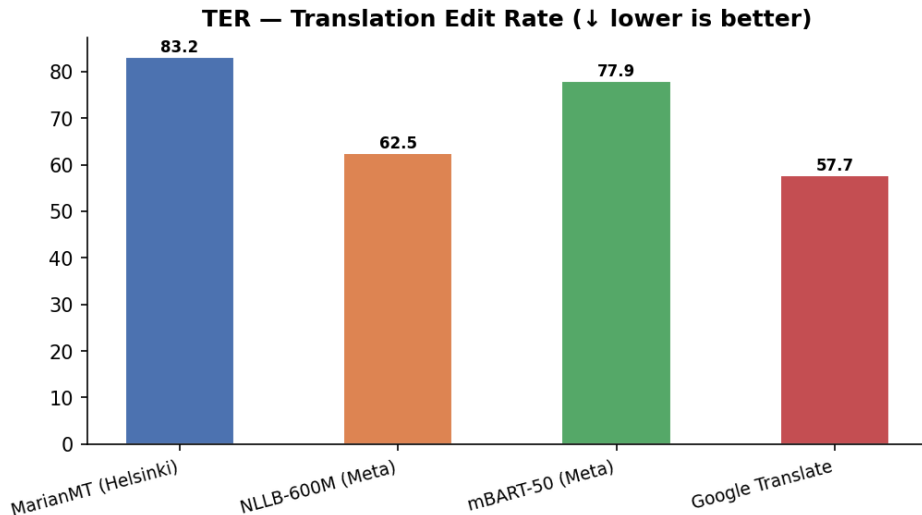
MarianMT BLEU is 24.5 points below Google Translate

Metrics Comparison Chart



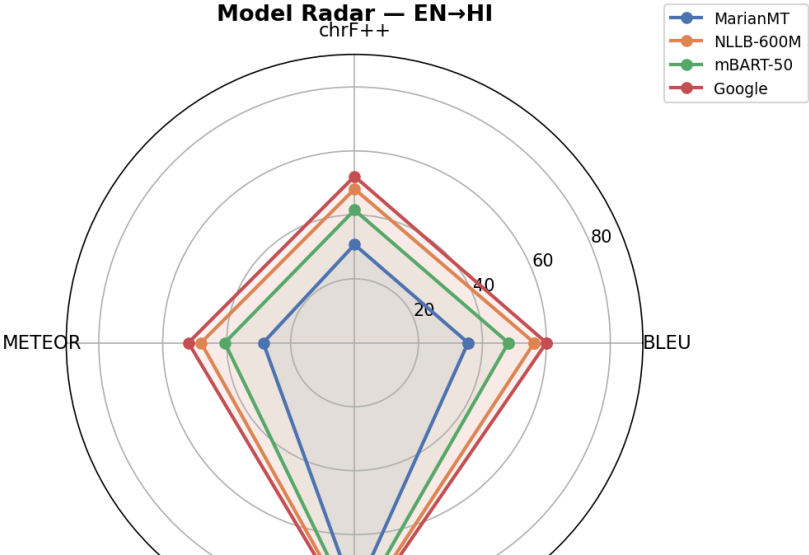
Bar chart of all five metrics across the four systems. Google Translate leads on all metrics; MarianMT significantly lags. NLLB-600M is the strongest open-source system.

TER Comparison



Translation Edit Rate (TER) — lower is better. Google Translate requires fewest edits (57.7) to

Radar Chart — Multi-Metric Profile



Rankings Summary

Model	B	C	T	M	BS	Avg
MarianMT	4	4	4	4	4	4.0
mBART-50	3	3	3	3	3	3.0
NLLB-600M	2	2	2	2	2	2.0
Google	1	1	1	1	1	1.0

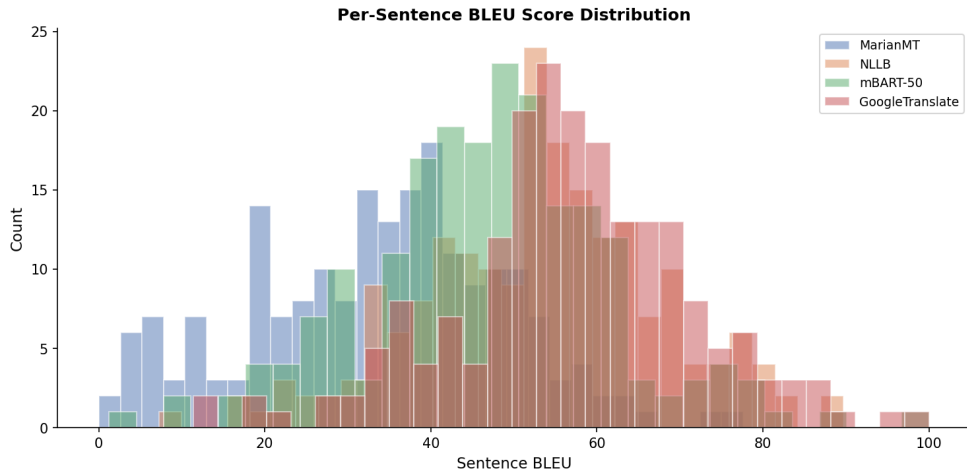
B=BLEU, C=chrF++, T=TER, M=METEOR,

BS=BERTScore

Key Observations

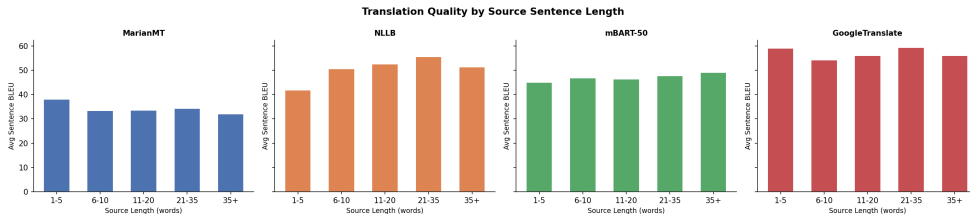
- Rankings are **perfectly consistent** across all metrics — no metric disagreement
- Google Translate's advantage is likely due to proprietary data scale
- NLLB-600M beats mBART-50 by **+8 BLEU** points
- MarianMT is **adequate as a baseline** but not production-ready for this pair

BLEU Score Distribution



Per-sentence BLEU score distributions. All models show right-skewed distributions with a long tail of low-scoring sentences, reflecting the inherent difficulty of certain sentence types.

Sentence Length vs. BLEU Score



BLEU score as a function of source sentence length. Short sentences (1–5 tokens) generally yield higher BLEU, while longer sentences (21+ tokens) trend lower due to compounding translation errors.

Hardest & Easiest Sentences

5 Hardest Sentences (avg BLEU)

- *"The water rail was fit and well by the time it was released."* — Idiomatic, rare vocabulary
- *"The rep adds: 'Perhaps she is confusing him...' "* — Quotation within quotation
- *"Kaplowitz contends the premise that holds the most weight is the epidemic of obesity."* — Complex nominal structure
- *"They never believed he died the way the sheriff concluded."* — Relative clause, named entity
- *"Court blocks ruling on NYPD stop-and-frisk policy"* — Headline, named entities

5 Easiest Sentences (avg BLEU)

- *"He did not tell anyone that Jahid has drowned."* — Simple, high-frequency vocabulary
- *"Today he will meet the families of the people who died in Patna last Sunday."* — Direct news style
- *"A southern Georgia judge on Wednesday ordered authorities to release all surveillance video."* — Factual
- *"He said: 'I really want to develop an invisible ice cream.' "* — Simple quoted speech
- *"They asked some subjects to count the number of dots on a computer screen."* — Simple active sentence

Common Translation Error Patterns

MarianMT Failure Modes

- Named entity hallucination / mistransliteration
- Incorrect script mixing (Latin chars in output)
- Dropped content words in long sentences
- Poor handling of numbers and dates

mBART-50 Issues

- Occasional word-order errors
- Struggles with domain-specific terms

NLLB-600M Strengths

- Best open-source fluency
- Handles Hinglish / code-mix better
- Consistent across length buckets

Sentence Length Effect

Across **all** models, BLEU degrades with sentence length:

- Short (1–5 tokens): avg $\sim 65+$ BLEU
- Medium (11–20 tokens): avg ~ 50 BLEU
- Long (21–35 tokens): avg ~ 40 –45 BLEU

Conclusions

Main Findings

- 1 **Google Translate** achieves the best performance across all five metrics (BLEU 60.0, BERTScore 87.25), likely due to proprietary data scale and continuous updates
- 2 **NLLB-600M** is the best open-source alternative (BLEU 56.2) and viable for production use cases
- 3 **mBART-50** occupies a middle ground — acceptable but beaten clearly by NLLB
- 4 **MarianMT** provides a useful, lightweight research baseline but is not competitive for high-quality EN→HI translation (BLEU 35.5)
- 5 All models struggle with **idiomatic**

Recommendations

- Use **Google Translate API** when accuracy is critical
- Use **NLLB-600M** for offline, privacy-sensitive, or cost-sensitive deployments
- Consider **domain fine-tuning** of NLLB for specialised corpora (medical, legal)
- Pre-process long sentences with sentence splitting to improve performance

Summary Dashboard

Model	BLEU	chrF++	TER	METEOR	BERTScore	Rank
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mBART-50	48.12	41.53	77.91	40.60	83.86	3 rd
NLLB-600M	56.20	48.11	62.52	48.02	85.85	2 nd
Google Translate	60.00	52.07	57.70	51.97	87.25	1st

• **Overall Winner: Google Translate** — Best on all 5 metrics with avg rank 1.0 •

Thank You

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Data, figures, and metrics available in the accompanying repository